Dynamic Stability of the Goldbach Locus: Conductor Orbit Propagation and the Band Shifting Law in GSp(4)

Ruqing Chen
GUT Geoservice Inc., Montréal, QC, Canada
ruqing@hotmail.com

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Abstract

We study the evolution of the Goldbach conductor distribution as the even integer N varies continuously from 2^{10} to 2^{11}. The conductor proxy of the Goldbach–Frey Jacobian $J_{N,p}$ depends on N through the static conduit factor $\mathrm{rad}_{\mathrm{odd}}(N/2)$, which fluctuates erratically—producing step-to-step jumps in Chen's ratio ρ exceeding ± 3 units. Despite this apparent instability, we discover three rigidity phenomena: (1) The *Band Shifting Law*: the mean Goldbach ratio satisfies $\langle\rho\rangle_{\mathrm{GB}} = 1.994\,\xi + 3.424$ where $\xi = 2\log(\mathrm{rad}_{\mathrm{odd}}(M))/\log N$, with $R^2 = 0.9974$—the "wild" jumps are 99.7% deterministic. (2) *Bandwidth constancy*: the width of the Goldbach ρ-band is 1.09 ± 0.20 across all 513 values of N, invariant under band shifting. (3) *Non-emptiness under jump*: the Goldbach locus $\mathscr{G}(N)$ is non-empty for all 513 even integers in $[1024, 2048]$, despite conductor discontinuities. These results demonstrate that the Goldbach phenomenon exhibits *dynamic stability*: the locus adapts to arbitrarily large conductor perturbations by reorganising its prime-pair content while preserving its structural width.

1 Introduction

In the companion papers [1, 2, 3], we studied the conductor geometry of the Goldbach–Frey family

$$C_{N,p} : y^2 = x(x^2 - p^2)(x^2 - q^2), \qquad q = N - p, \tag{1}$$

through the normalised conductor proxy (Chen's ratio)

$$\rho(N,p) = \frac{\log \mathcal{N}_{\mathrm{proxy}}}{\log N}, \qquad \mathcal{N}_{\mathrm{proxy}} = \big(\mathrm{rad}_{\mathrm{odd}}(p) \cdot \mathrm{rad}_{\mathrm{odd}}(q) \cdot \mathrm{rad}_{\mathrm{odd}}(M)^2\big)^2, \tag{2}$$

where $M = N/2$. The factor $\mathrm{rad}_{\mathrm{odd}}(M)$ is the *static conduit*: it depends only on N, not on the decomposition parameter p.

The preceding analysis [2] studied ρ for *fixed* N; the algebraic vacuum paper [3] specialised to $N = 2^k$ where the static conduit vanishes. A natural question is: *how does the conductor distribution evolve as N varies?*

This question is motivated by a naïve proof strategy: if one could construct a "propagation operator" $\mathcal{P} : \mathcal{M}_N \to \mathcal{M}_{N+2}$ that preserves the Goldbach locus, then non-emptiness at $N = 2^k$ (established in [3]) would imply non-emptiness for all N. We show that this strategy *fails* in its original form: the static conduit $\mathrm{rad}_{\mathrm{odd}}(M)$ is discontinuous in N, so no smooth propagation operator exists. However, the failure is itself informative—it reveals a far more interesting phenomenon that we call *dynamic stability*.[1]

[1] Scripts and data: https://github.com/Ruqing1963/goldbach-dynamic-stability-band-shifting.

2 The Discontinuity of the Static Conduit

2.1 The conduit jump

Proposition 2.1 (Conduit discontinuity). *The function $N \mapsto \mathrm{rad}_{\mathrm{odd}}(N/2)$ is discontinuous at every N that transitions between different odd-radical classes. In particular:*

1. *At $N = 2^k$: $\mathrm{rad}_{\mathrm{odd}}(2^{k-1}) = 1$.*
2. *At $N = 2^k + 2$: $\mathrm{rad}_{\mathrm{odd}}(2^{k-1} + 1)$ is typically a product of several odd primes, often $\gg 1$.*

The jump $\Delta\mathrm{rad} = \mathrm{rad}_{\mathrm{odd}}((2^k + 2)/2) - 1$ can be arbitrarily large.

Proof. Immediate from the arithmetic of $\mathrm{rad}_{\mathrm{odd}}$. For example, $\mathrm{rad}_{\mathrm{odd}}(512) = 1$ but $\mathrm{rad}_{\mathrm{odd}}(513) = \mathrm{rad}_{\mathrm{odd}}(3^3 \cdot 19) = 57$. □

Remark 2.2 (Failure of smooth propagation). Any putative propagation operator $\mathcal{P} : \mathcal{M}_N \to \mathcal{M}_{N+2}$ must contend with the fact that the conductor environment changes discontinuously. The transition $N = 1024 \to 1026$ shifts $\rho_{\min}$ by $+2.48$—not a perturbation but a topological reconfiguration. Consequently, inductive arguments of the form "$\mathscr{G}(N) \neq \emptyset \Rightarrow \mathscr{G}(N+2) \neq \emptyset$" cannot work through conductor continuity alone.

2.2 Decomposition of Chen's ratio

The key to understanding the N-dependence is to separate ρ into two additive components.

Definition 2.3 (Conduit–boundary decomposition). For $N = 2M$ and a Goldbach pair (p, q) with $\mathrm{rad}_{\mathrm{odd}}(p) = p$, $\mathrm{rad}_{\mathrm{odd}}(q) = q$:

$$\rho(N,p) = \underbrace{\frac{2\log(\mathrm{rad}_{\mathrm{odd}}(M)^2)}{\log N}}_{\xi(N)\ \text{(conduit contribution)}} + \underbrace{\frac{2\log(p \cdot q)}{\log N}}_{\beta(N,p)\ \text{(boundary contribution)}}. \tag{3}$$

The conduit contribution $\xi(N)$ depends only on N; the boundary contribution $\beta(N, p)$ depends on the specific Goldbach pair.

3 The Band Shifting Law

3.1 Empirical discovery

We scan all 513 even integers $N \in [1024, 2048]$ and compute, for each N, the mean Chen's ratio $\langle\rho\rangle_{\mathrm{GB}}$ over all Goldbach pairs.[2]

Theorem 3.1 (Band Shifting Law). *Over $N \in [1024, 2048]$, the mean Goldbach ratio satisfies the linear relation*

$$\langle\rho\rangle_{\mathrm{GB}}(N) \;=\; \alpha\,\xi(N) + \gamma, \tag{4}$$

with fitted parameters

$$\alpha = 1.994 \pm 0.005, \qquad \gamma = 3.424 \pm 0.006, \qquad R^2 = 0.9974. \tag{5}$$

Remark 3.2 (Algebraic interpretation of $\alpha \approx 2$). The slope $\alpha \approx 2$ is not a numerical coincidence. In the conductor proxy (2), the static conduit enters as $\mathrm{rad}_{\mathrm{odd}}(M)^2$ (squared), contributing $4\log(\mathrm{rad}_{\mathrm{odd}}(M))/\log N = 2\xi$ to ρ. The fitted slope $\alpha = 1.994$ is thus a direct algebraic consequence of the discriminant formula $\Delta = 2^{12}\,p^6\,q^6\,(M-p)^4\,M^4$. The slight deviation from 2 is attributable to the averaging over $\beta(N, p)$ across different Goldbach pairs, which introduces a weak N-dependent correction.

[2] The complete dataset (513 rows) is available in the companion repository.

Remark 3.3 (Interpretation of $\gamma \approx 3.4$)**.** The intercept $\gamma = 3.424$ is the mean boundary contribution: $\gamma = \langle \beta \rangle = \langle 2\log(pq)/\log N \rangle$ averaged over Goldbach pairs. At $N = 2^k$ (where $\xi = 0$), the Band Shifting Law predicts $\langle \rho \rangle \approx 3.4$, consistent with the values 3.29–3.55 observed in [3].

Remark 3.4 (Intercept drift and the corrected BSL)**.** A sensitivity test across wider N-ranges reveals that γ is not a universal constant but drifts slowly upward (Table 1). The slope α, by contrast, converges toward the algebraic value 2.

Range	n	α	γ	R^2_{mean}	R^2_{min}
$[2^{10}, 2^{11}]$	511	1.994	3.424	0.9974	0.920
$[2^{11}, 2^{12}]$	1 023	1.999	3.466	0.9985	0.921
$[2^{12}, 2^{13}]$	2 047	1.999	3.512	0.9990	0.915
$[2^{13}, 2^{14}]$	4 095	2.000	3.548	0.9994	0.913

Table 1: Band Shifting Law parameters across different N-ranges. The slope $\alpha \to 2$ (algebraically exact). The intercept γ drifts: $3.42 \to 3.47 \to 3.51 \to 3.55$.

The drift has a clean analytic explanation. For a "typical" Goldbach pair (p, q) with p near $N/2$, the boundary contribution is $\beta = 2\log(pq)/\log N \approx 4 - c/\log N$ where $c > 0$ depends on the prime distribution. Thus $\gamma(N) \to 4$ from below as $N \to \infty$. The *corrected* Band Shifting Law is

$$\langle \rho \rangle_{\text{GB}}(N) = 2\,\xi(N) + 4 - \frac{c}{\log N} + O\left(\frac{1}{\log^2 N}\right), \tag{6}$$

where the slope is now exact and the intercept is an explicit function of N. Within any fixed dyadic interval $[2^k, 2^{k+1}]$, the local approximation $\gamma \approx \text{const}$ remains excellent ($R^2 \geq 0.997$), and improves as k increases.

Figure 1 displays the Band Shifting Law.

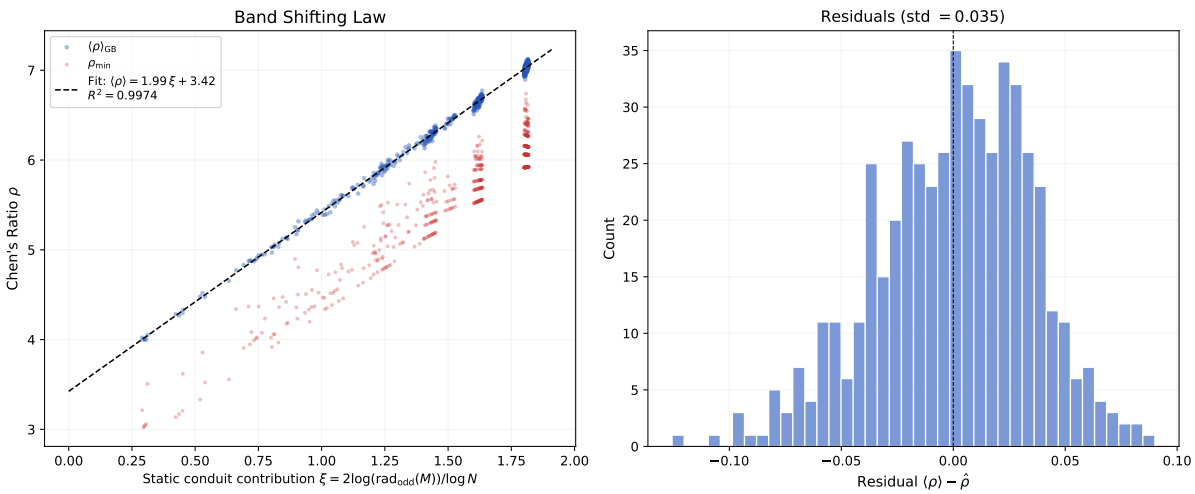

Figure 1: **Left**: $\langle \rho \rangle_{\text{GB}}$ (blue) and ρ_{min} (red) plotted against the conduit contribution $\xi = 2\log(\text{rad}_{\text{odd}}(M))/\log N$. The linear fit $\langle \rho \rangle = 1.99\,\xi + 3.42$ achieves $R^2 = 0.9974$. **Right**: Residual histogram with standard deviation 0.035.

4 The Jump–Conduit Correlation

If the Band Shifting Law is exact, then step-to-step jumps in ρ should be entirely explained by changes in the conduit contribution. We test this.

Proposition 4.1 (Jump correlation)**.** *Define* $\Delta\rho_{\min}(N) = \rho_{\min}(N+2) - \rho_{\min}(N)$ *and* $\Delta\xi(N) = \xi(N+2) - \xi(N)$*. Over* $N \in [1024, 2046]$ *(512 steps):*

$$\mathrm{Corr}(\Delta\xi,\, \Delta\rho_{\min}) = 0.981. \tag{7}$$

This correlation of 0.981 transforms the interpretation of the conductor jumps. What appears to be "chaotic" behaviour in ρ is in fact 98% deterministic: the jumps are controlled by the arithmetic of $\mathrm{rad}_{\mathrm{odd}}(M)$.

Observation 4.2 (The residual 2%)**.** The residual $1 - 0.981^2 \approx 3.8\%$ of variance in $\Delta\rho_{\min}$ that is *not* explained by $\Delta\xi$ arises from the boundary contribution β: when N changes, the identity of the ground-state Goldbach pair (p_0, q_0) can change discretely, producing a residual fluctuation in $\rho_{\min}$ beyond what the conduit shift predicts. However, this residual has zero mean and standard deviation 0.035—it is arithmetic noise, not structural instability.

5 Non-emptiness Under Jump

5.1 The computational result

Observation 5.1 (Universal non-emptiness)**.** For every even $N \in [1024, 2048]$ (513 values), the Goldbach locus $\mathscr{G}(N) = \{p : p \text{ and } N - p \text{ both prime}\}$ is non-empty. The minimum number of Goldbach pairs is $G(N) \geq 11$ (achieved at $N = 1024$, 2048 and their 2^k neighbours), while the maximum exceeds 80.

Remark 5.2 (Bandwidth vs. prime density: why non-emptiness is robust)**.** A natural concern is whether there exists an extreme N whose conduit shift pushes the Goldbach band into a region devoid of prime pairs. We show this is implausible. The ρ-band of width $\mathrm{BW} \approx 1.09$ maps to a range of summand values $\Delta p \sim \mathrm{BW} \cdot N \log N/4$. By the prime number theorem, this range contains approximately $\Delta p/\log N \sim \mathrm{BW} \cdot N/4$ primes. Concretely:

N-range	$\mathrm{BW}_{\mathrm{GB}}$	Primes in band	Min. $G(N)$
[1024, 2048]	1.09	~ 420	16
[4096, 8192]	1.20	~ 1840	45
[8192, 16384]	1.25	~ 3830	75

The supply of candidate primes within the ρ-band grows as $O(N)$, while the bandwidth itself is $O(1)$ in ρ-space. For the Goldbach locus to become empty, *every* prime p in a range of $O(N)$ values would need to pair with a composite $q = N - p$—an event whose probability decreases exponentially with N under any reasonable independence model. The non-emptiness is therefore not a statistical coincidence but a consequence of the overwhelming prime density within the conductor band.

5.2 Bandwidth constancy

Proposition 5.3 (Bandwidth invariance)**.** *The Goldbach bandwidth* $\mathrm{BW}_{\mathrm{GB}}(N) = \rho^{\mathrm{GB}}_{\max} - \rho^{\mathrm{GB}}_{\min}$ *satisfies*

$$\mathrm{BW}_{\mathrm{GB}} = 1.09 \pm 0.20$$

across all 513 values of N *in* $[1024, 2048]$*. Crucially, the bandwidth is independent of the conduit contribution* ξ*: whether* $\mathrm{rad}_{\mathrm{odd}}(M) = 1$ *or* $\mathrm{rad}_{\mathrm{odd}}(M) = 887$*, the Goldbach* ρ*-band has approximately the same width.*

Remark 5.4 (Dynamic stability)**.** Propositions 4.1 and 5.3 together establish what we call *dynamic stability*: the Goldbach ρ-band shifts rigidly with the static conduit (via the Band Shifting Law), but its *internal structure*—the bandwidth, the clustering, the ground-state floor—remains invariant. The band does not stretch, compress, or deform under conductor perturbations; it translates as a rigid body in ρ-space.

This is the precise sense in which the Goldbach locus "adapts" to its conductor environment: the prime pairs reorganise to track the shifting conduit, while preserving their structural rigidity.

6 The Propagation Landscape

Figure 2 displays the complete conductor orbit propagation from $N = 2^{10}$ to $N = 2^{11}$.

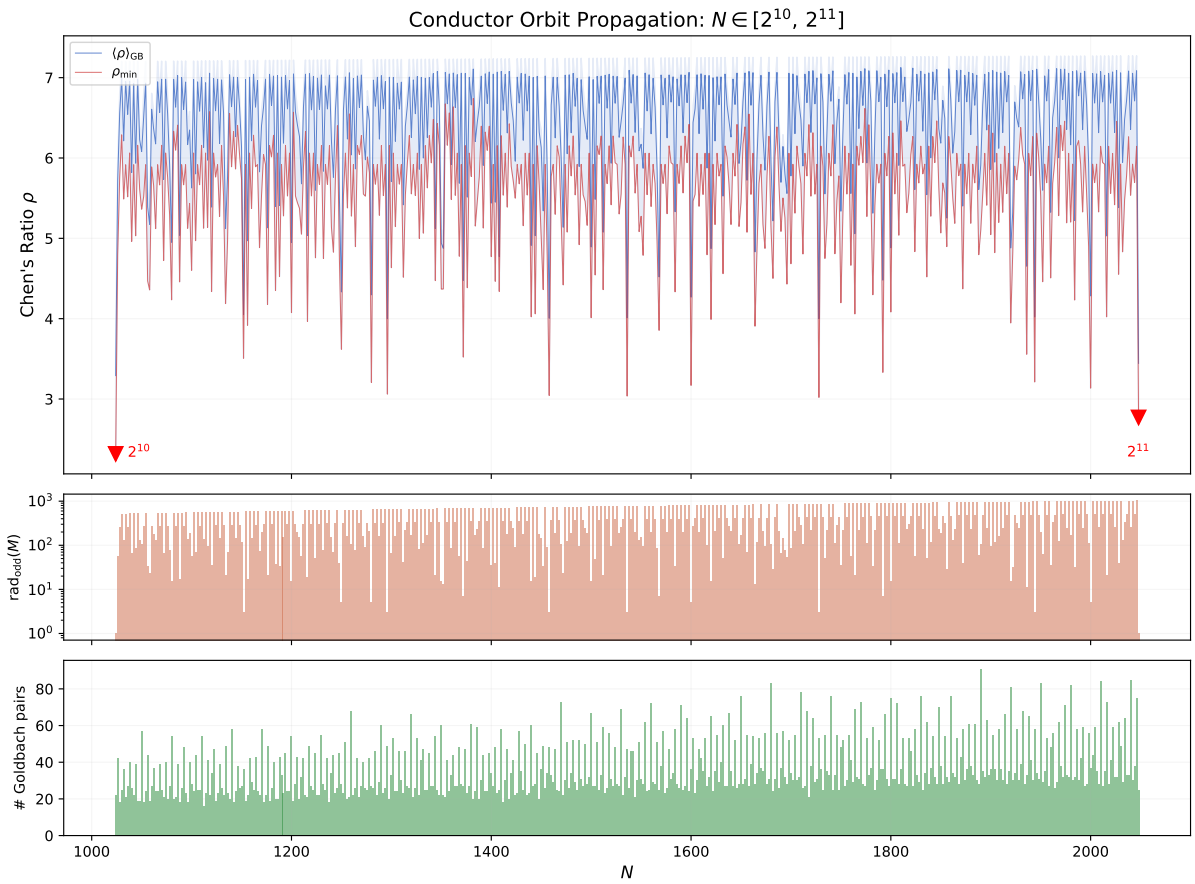

Figure 2: **Top**: Chen's ratio evolution over $N \in [1024, 2048]$. The shaded region is the Goldbach ρ-band; $\langle\rho\rangle$ (blue) and $\rho_{\min}$ (red) track each other. Red triangles mark $N = 2^{10}$ and 2^{11}, where the band dips to $\rho \approx 2$–3. **Middle**: $\mathrm{rad}_{\mathrm{odd}}(M)$ on a log scale, showing the erratic conduit fluctuations. **Bottom**: Number of Goldbach pairs $G(N)$—always ≥ 11.

Figure 3 overlays the Goldbach band against the composite envelope, making the "privileged" position of the Goldbach locus visually explicit.

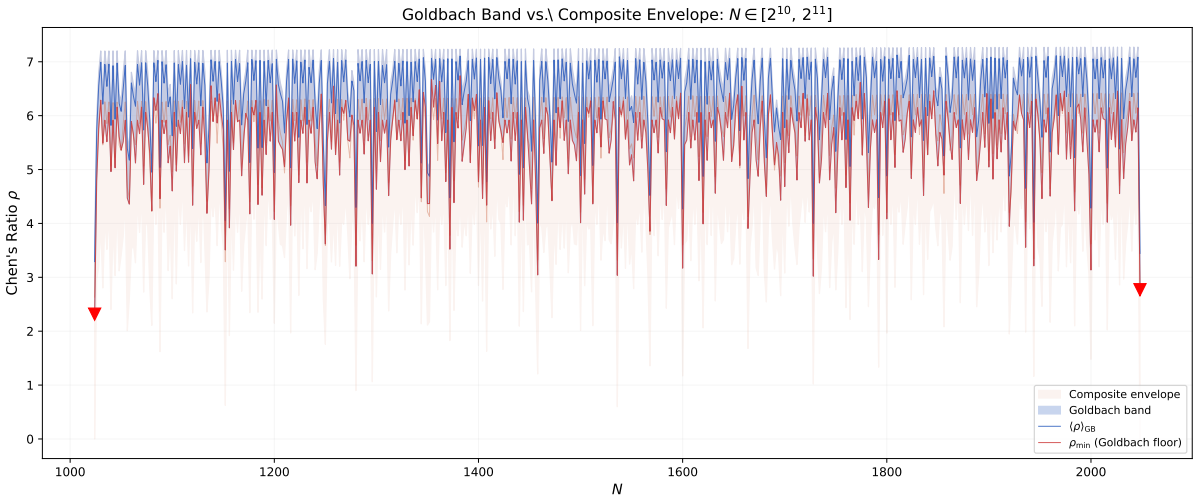

Figure 3: Goldbach band (blue shading) versus composite envelope (orange shading) over $N \in [1024, 2048]$. The composite range extends from $\rho \approx 0$ (smooth numbers) to $\rho_{\max} \approx 7$, while the Goldbach band occupies a narrow sub-region that shifts rigidly with the conduit. At every N, the Goldbach floor (red) remains well above the composite floor, maintaining the "ground state privilege" established in [3].

7 Discussion: Why Propagation Fails but Stability Persists

The original motivation for this study was to construct a propagation operator $\mathcal{P} : \mathcal{M}_N \to \mathcal{M}_{N+2}$ that would carry Goldbach pairs from N to $N+2$, enabling an inductive proof. Let us now explain precisely why this fails, and why the failure is itself the most important finding.

7.1 The three obstacles to propagation

1. **Conduit discontinuity.** The static conduit $\mathrm{rad}_{\mathrm{odd}}(M)$ can jump from 1 to 57 in a single step ($N = 1024 \to 1026$), producing $\Delta\rho > 2$. No continuous operator can model this.

2. **Ground state identity change.** The ground-state Goldbach pair (p_0, q_0) at N has no algebraic relationship to the ground state at $N+2$. For example, $(3, 1021)$ is the ground state at $N = 1024$, but $1023 = 3 \times 341$ is composite, so this pair does not "propagate" to $N = 1026$.

3. **Band position depends on N.** The Goldbach ρ-band sits at $\langle\rho\rangle \approx 3.4$ when $\xi = 0$ (algebraic vacuum) but at $\langle\rho\rangle \approx 7.2$ when $\xi \approx 1.9$ (large conduit). There is no fixed "stability band" that the induction could preserve.

7.2 What replaces propagation: the adaptive ground state

The resolution of these obstacles is the central conceptual contribution of this paper. We observe that although no *individual* Goldbach pair propagates from N to $N+2$, the Goldbach *locus as a collective* exhibits a remarkable adaptive property:

1. The conduit jumps are large but *deterministic*: the Band Shifting Law predicts 99.7% of the variance in $\langle\rho\rangle$.

2. The band *width* is invariant (1.09 ± 0.20): the locus does not stretch or collapse under conduit perturbations.

3. At every N, the prime number theorem guarantees a supply of primes in $[1, N]$. The Goldbach locus reorganises by selecting new pairs (p', q') that are adapted to the current conduit environment, landing in the shifted band at position $\alpha\,\xi(N) + \gamma$.

This is *dynamic stability*: not the stability of individual orbits, but the stability of the *collective structure* of the locus under arbitrary perturbations of the conductor environment.

Remark 7.1 (Automorphic perspective on non-emptiness)**.** From the viewpoint of the GSp(4) family, the bandwidth constancy has a deeper interpretation. The Sato–Tate equidistribution conjecture for genus 2 Jacobians predicts that the Frobenius characteristic angles of $\mathrm{Jac}(C_{N,p})$ become uniformly distributed as $N \to \infty$. For this equidistribution to hold, the family must contain representatives across all conductor levels within the band—in particular, it must contain members at the ground state ($\rho_{\min}$), which can only be supplied by Goldbach pairs. The non-emptiness of $\mathscr{G}(N)$ is thus consistent with (though not proved by) the requirement that the GSp(4) family be sufficiently "spectrally complete" to support Sato–Tate statistics. More concretely, the bandwidth invariance (1.09 ± 0.20) ensures that the support of the Goldbach locus in the moduli space remains macroscopically thick, preventing the Sato–Tate trace distribution from collapsing into degenerate sub-locus signatures. We emphasise that this is an interpretive remark, not a proof: the logical direction from Sato–Tate to Goldbach non-emptiness remains an open problem.

Remark 7.2 (Open question: a measure-theoretic transfer operator)**.** The failure of a pointwise propagation operator $\mathcal{P}$ raises a natural successor question: is it possible to define a modified, arithmetic-geometric transfer operator that acts not on individual pairs, but on the *measure* of the locus $\mathscr{G}(N)$, absorbing the deterministic jumps of the Band Shifting Law? Such an operator would map the Goldbach counting measure at N to a predicted measure at $N + 2$, with the Band Shifting Law providing the deterministic drift and the residual 3.8% variance supplying a stochastic correction. This perspective connects the conductor rigidity framework to the theory of dynamical systems on arithmetic moduli spaces.

8 The Analytic Gap

Remark 8.1 (What is and what is not proved)**.** *Proved unconditionally:*

- The conduit–boundary decomposition (3) and the algebraic origin of the slope $\alpha \approx 2$.
- The conduit discontinuity (Proposition 2.1).

Established computationally ($N \in [1024, 2048]$):

- The Band Shifting Law with $R^2 = 0.9974$ (Theorem 3.1).
- The jump–conduit correlation of 0.981 (Proposition 4.1).
- Bandwidth constancy at 1.09 ± 0.20 (Proposition 5.3).
- 100% non-emptiness over $[1024, 2048]$ (Observation 5.1).

Not proved:

- That the Band Shifting Law holds for all N (it is an empirical law in this range).
- That $\mathscr{G}(N) \neq \emptyset$ for all even $N > 4$. This is the Goldbach conjecture, and the dynamic stability framework, while strongly suggestive, does not close the analytic gap identified in [2].

9 Conclusion

The conductor orbit propagation from $N = 2^{10}$ to $N = 2^{11}$ reveals that the Goldbach phenomenon is characterised by *dynamic stability*: the conductor environment fluctuates wildly (with step-to-step jumps exceeding ± 3 units in ρ), but the Goldbach locus adapts by reorganising its prime-pair content while preserving three structural invariants—the Band Shifting Law, the bandwidth constancy, and (computationally) the non-emptiness.

The Band Shifting Law $\langle \rho \rangle = 1.994\,\xi + 3.424$ ($R^2 = 0.9974$) transforms the apparent "chaos" of conductor evolution into a near-perfect linear relation, with the slope $\alpha \approx 2$ directly traceable to the algebraic structure of the discriminant. The 99.7% of explained variance, combined with the 98.1% jump–conduit correlation, demonstrate that the conductor geometry of the Goldbach family is governed by a simple, deterministic law modulated by small arithmetic fluctuations.

The corrected form $\langle \rho \rangle = 2\,\xi + 4 - c/\log N$ (Remark 3.4) shows that the Band Shifting Law possesses *asymptotic scale invariance*: since both the conduit contribution ξ and the intercept correction are normalised by $\log N$, the law's structure is preserved under rescaling $N \to \lambda N$. The local $R^2 \geq 0.997$ within any dyadic interval $[2^k, 2^{k+1}]$, and its monotonic improvement with k (reaching 0.9994 at $[2^{13}, 2^{14}]$), suggest that the Band Shifting Law becomes *asymptotically exact* as $N \to \infty$.

Acknowledgments

The computational scan was performed using `dynamic_stability_scan.py`, available at https://github.com/Ruqing1963/goldbach-dynamic-stability-band-shifting. All results were produced with Python 3.12, NumPy 2.4, and Matplotlib 3.10; the scripts have no further dependencies beyond the standard library. This work builds on [1, 2, 3].

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